Title:

Downside Tolerance and Drawdown Absorption in a Depth-Controlled Spot Trading Strategy: Empirical Evidence from Cryptocurrency Markets

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Abstract

Martingale-based trading systems are commonly criticized for their tendency toward unbounded exposure growth and catastrophic failure during adverse price movements. Such assessments, however, are largely derived from static doubling schemes or short-horizon performance evaluations, offering limited insight into how execution-level design constraints influence system behavior under prolonged market stress.

This study examines the survivability characteristics of a depth-controlled Martingale-based spot trading system by analyzing its realized execution paths during extended adverse market regimes. Rather than proposing a new trading framework, the analysis focuses on how an existing depth-constrained system operates under multi-month downside conditions characterized by repeated breakdowns, partial recoveries, and sustained execution pressure.

Using reconstructed trade-level data from cryptocurrency spot markets, we investigate the evolution of exposure depth, capital utilization, and execution pacing along realized price trajectories. System evaluation emphasizes operational continuity, containment of localized failures, and the ability to transition back to stable execution states, rather than aggregate profitability or drawdown magnitude.

The findings suggest that, within clearly defined structural limits, depth-controlled Martingale systems can function as downside absorption mechanisms, transforming extended adverse price movements into a sequence of bounded execution states. The analysis further identifies explicit failure boundaries—including depth saturation, execution overload, liquidity deterioration, and behavioral discipline erosion—beyond which system survivability deteriorates rapidly.

By shifting the evaluation of Martingale-derived strategies from model-centric optimality toward realized stress-path behavior, this study contributes an execution-level perspective on risk management in non-stationary markets.

Keywords:
Martingale trading; downside risk; execution-level analysis; drawdown tolerance; cryptocurrency spot markets; survivability

1. Introduction

Martingale-based trading strategies are widely regarded as structurally fragile due to their inherent tendency toward escalating exposure and severe drawdown risk during adverse price movements. Empirical and survey-based studies have consistently shown that drawdowns capture dimensions of downside risk that are not reflected by variance-based or expectation-based measures, particularly under prolonged unfavorable market conditions (Choi, 2021; Geboers et al., 2023). As a result, Martingale-derived systems are often characterized as pathological risk processes rather than viable components of systematic trading frameworks.

These concerns are further amplified in non-stationary financial environments characterized by regime shifts, structural breaks, and evolving volatility dynamics. Recent evidence suggests that conventional performance and risk evaluation frameworks lose explanatory power when market dynamics deviate from stationarity assumptions (Cappelli et al., 2021; Varga & Szendrei, 2025). Under such conditions, drawdown behavior becomes increasingly path-dependent, rendering static performance metrics insufficient for assessing system robustness.

Despite the extensive literature on drawdown risk, much of the existing research evaluates trading systems through terminal outcomes—such as maximum drawdown or recovery time—while abstracting away execution-level dynamics. In particular, limited attention has been paid to how capital deployment, execution pacing, and exposure depth evolve along realized price paths during sustained stress episodes (Bernard et al., 2021; Zhang & Li, 2023). This outcome-centric perspective obscures the operational processes through which risk accumulates and is potentially contained.

This study argues that evaluating Martingale-based systems solely through aggregate risk metrics overlooks a critical dimension: survivability under continuous execution pressure. Rather than assessing whether such systems are optimal or profitable, we examine whether they can maintain operational continuity when exposed to extended downside regimes. Recent research highlights the importance of execution-path analysis and stress dynamics in understanding the resilience of algorithmic trading systems (D'Amico et al., 2024; Wang & Xi, 2025).

Using reconstructed trade-level records from cryptocurrency spot markets, this study analyzes a depth-controlled Martingale system subjected to a drawdown exceeding 40%. The system employs explicit depth constraints and non-doubling position increments, allowing exposure growth to remain bounded within a predefined structural domain. The results provide empirical evidence that, within such boundaries, drawdown can be transformed into a sequence of manageable execution states rather than a single catastrophic failure. At the same time, we explicitly identify failure boundaries—such as depth saturation, execution overload, liquidity deterioration, and discipline erosion—beyond which survivability deteriorates rapidly (Dombrowski et al., 2023; Benjlijel & Mansali, 2021).

2. Strategy Overview and Execution Environment

2.1 Analytical Scope and Interpretation

The trading framework analyzed in this study is not introduced as a novel or optimized strategy design. Instead, it is examined as an *execution mechanism* operating within a predefined structural domain. The purpose of this section is to clarify the operational context under which the observed trade paths are generated, rather than to prescribe a replicable trading methodology.

Accordingly, the strategy description should be interpreted as a characterization of how a depth-controlled Martingale mechanism behaves under sustained adverse price dynamics, given explicit execution constraints and platform-imposed design boundaries.

2.2 Asset Selection and Market Context

The empirical analysis focuses on ETH traded in the spot market. ETH was selected not for its investment appeal, but for its suitability as a stress-testing asset. As one of the most liquid digital assets, ETH exhibits continuous trading, deep order books, and prolonged market history across multiple volatility regimes. At the same time, historical evidence shows that ETH is capable of experiencing extended drawdowns exceeding 30–50% without market discontinuity, abrupt delisting risk, or liquidity collapse.

These characteristics make ETH particularly appropriate for examining execution-level survivability under sustained downside pressure. The asset provides a realistic environment in which drawdown dynamics unfold gradually over time, allowing exposure depth, execution pacing, and capital utilization to be observed under genuine market stress rather than isolated shock events.

2.3 Platform-Imposed Execution Constraints

The execution framework is implemented using standard functionality provided by a centralized cryptocurrency exchange. This includes features such as trailing

Martingale-style accumulation and indicator-based entry gating. These mechanisms are not custom-designed by the author, nor are they introduced to evaluate the effectiveness of specific indicators or parameter settings.

Instead, the observed execution behavior reflects the *behavioral space defined by the trading platform itself*. Users interacting with such systems operate within a constrained configuration domain determined by exchange-provided tools, default parameter ranges, and fixed execution rules. As a result, the realized trade paths analyzed in this study arise from platform-mediated constraints rather than from freely optimized or bespoke strategy construction.

This distinction is critical for interpretation. The analysis does not seek to endorse the use of particular indicators (e.g., RSI thresholds) or Martingale variants, but rather to examine how a Martingale-type mechanism—*as it is practically accessible to market participants*—responds to prolonged downside regimes.

2.4 Structural Characteristics of the Execution Framework

Within the platform-defined environment, the execution mechanism exhibits the following structural features:

- **Micro-position decomposition:**
  Capital is allocated across multiple small tranches rather than being deployed through geometric or doubling position increments.
- **Explicit maximum depth constraint:**
  The execution process operates within a predefined depth boundary, beyond which no further averaging or exposure expansion occurs.
- **Event-driven, non-continuous execution:**
  Entries are conditionally triggered based on market states rather than deployed through dense or grid-saturated placement.
- **Spot-only exposure with no leverage:**
  All positions are held in spot markets, eliminating forced liquidation risk associated with margin or derivative products.

- **Cost-basis compression objective:**
  The execution logic is oriented toward controlled reduction of effective entry cost during adverse price movements, rather than toward short-term price prediction.

These characteristics collectively define a bounded execution domain in which exposure growth is structurally limited and temporally paced.

2.5 Operational Objective: Survivability over Optimality

Crucially, the execution framework is not designed to maximize hit rates, short-term profitability, or conventional risk-adjusted performance metrics. Its primary operational objective is to remain structurally intact under adverse price paths.

The system accepts low-frequency execution, extended holding periods, and prolonged drawdown exposure as inherent features of its design. From an analytical perspective, this allows the system to be examined as a *downside absorption mechanism*—one that transforms sustained adverse price movements into a sequence of manageable execution states, rather than a single terminal failure.

This framing aligns with the survivability-oriented perspective articulated in the Introduction, emphasizing execution continuity and failure containment over outcome-centric performance evaluation.

2.6 Implications for Empirical Analysis

Given the platform-imposed constraints and the non-prescriptive nature of the execution framework, the empirical analysis that follows focuses on *realized execution paths* rather than theoretical optimality. The study evaluates how exposure depth, capital utilization, and execution states evolve during an extended drawdown episode exceeding 40%, without attributing causal significance to specific indicator signals or parameter choices.

Accordingly, the findings should be interpreted as evidence of how a depth-controlled Martingale mechanism—operating within realistic exchange-level constraints—behaves under prolonged stress, rather than as validation of a particular trading strategy.

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3. The 42% Drawdown Event

This section examines a prolonged downside stress episode in the ETH spot market, hereafter referred to as the *42% drawdown event*, and its execution-level implications for the analyzed strategy. Between August 25, 2025, and December 18, 2025, the ETH spot price declined from a local high near 4,950 USDT to a trough around 2,830 USDT, corresponding to a cumulative drawdown of approximately 42.8%. As illustrated in Figure 2, this movement did not manifest as a short-lived volatility spike or a singular liquidation cascade. Instead, it unfolded as a sustained, multi-month decline characterized by persistent downside pressure, intermittent relief rallies, and repeated violations of local support levels.

While Figure 2 documents the external market price trajectory during the drawdown shock, Figure 1 reports the cumulative execution-level performance of the ETH spot tracking-Martingale system over its full live operation period (approximately 800 days). The shaded region in Figure 1 corresponds to the same drawdown interval highlighted in Figure 2. Taken together, the two figures provide a synchronized, time-aligned view of market stress and system behavior, allowing direct examination of how a prolonged downside shock translated into sustained execution demands rather than abrupt terminal failure.

From an execution-level perspective, this interval functioned as a real-market stress exposure rather than a stylized statistical tail event. The drawdown continuously imposed pressure on position depth, capital allocation, and execution pacing, compelling the strategy to operate under conditions that conventional financial models would typically classify as hostile or terminal. Importantly, this stress environment was neither simulated nor temporally compressed. It unfolded under live market conditions, with full exposure to liquidity fluctuations, execution frictions, and the cumulative psychological load inherent in extended drawdown regimes.

Within this context, the 42.8% ETH decline serves less as a performance datapoint and more as a *shock-absorption test*, revealing how the system behaves once theoretical assumptions such as stationarity, distributional stability, and smooth mean reversion no longer apply. Survivability in this episode was not inferred from expected values or probabilistic forecasts, but enforced through time, path dependence, and continuous execution demands.

From the perspective of classical Martingale risk theory, a drawdown of this magnitude is typically regarded as structurally fatal. In leveraged or unconstrained averaging systems, such a movement would almost certainly result in forced liquidation or irreversible capital impairment. In contrast, during this episode the analyzed strategy remained fully operational throughout the entire drawdown period. No liquidation occurred, no emergency capital injection was required, and exposure remained explicitly bounded through predefined depth constraints and controlled execution pacing.

As the drawdown progressed, the effective cost basis was progressively compressed rather than destabilized, allowing subsequent price stabilization to enable partial and eventual profit realization without requiring a structural reset of the system. In this sense, the drawdown was *absorbed rather than resisted*. The strategy did not attempt to predict or negate the decline; instead, it responded to adverse price paths through pre-engineered execution rules designed to preserve operational continuity under prolonged stress.

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4. Interpretation: Martingale as an Absorption System

This episode highlights a critical but often under-discussed distinction in the interpretation of Martingale-based strategies. Martingale systems do not inherently fail because prices decline; they fail when depth and capital allocation are mis-engineered. When exposure growth is unconstrained or improperly paced, drawdowns become destabilizing. When depth and allocation are explicitly bounded, the same price movements can instead be absorbed.

Within the present framework, drawdowns are not interpreted as signals of failure, but as structural inputs to the system. Each incremental price decline activates a bounded response that limits marginal exposure while simultaneously improving the system's capacity to benefit from even modest rebounds. This behavior has not been inferred from a single event but observed repeatedly over time under the current operating conditions.

Importantly, the absorption process does not occur in a single, continuous motion. During the extended drawdown period, the strategy realized profits on multiple occasions through partial take-profit events, while also repeatedly re-engaging during subsequent downside movements. These cycles of partial realization and re-entry effectively dissipated downside pressure in stages, rather than allowing it to accumulate unchecked. In this sense, the drawdown was not merely endured but incrementally processed.

This repeated pattern suggests that, at least at certain execution levels, Martingale-based mechanisms can function as effective tools for managing downside pressure. Rather than amplifying risk mechanically, the strategy redistributed exposure across depth in a controlled manner, allowing portions of the accumulated drawdown to be relieved through intermittent price recoveries.

From this perspective, the Martingale mechanism functions less like a gambling progression and more like a nonlinear buffer, reallocating exposure across depth rather than time. Its role is not to predict reversals or eliminate losses, but to transform extended adverse price paths into a sequence of manageable execution states. While this does not render the strategy universally robust, it does indicate that, under appropriate structural constraints, Martingale-derived systems can meaningfully address the problem of sustained downside pressure.

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5. Where Absorption Breaks Down

While the preceding analysis illustrates how a depth-controlled Martingale framework can function as a downside absorption system under sustained drawdowns, it is equally important to clarify the conditions under which this absorption mechanism fails. The capacity to absorb adverse price movement is not unlimited, nor is it invariant across regimes. Absorption is a structural property, not a guarantee.

First, absorption breaks down when depth constraints are violated. The mechanism described here relies fundamentally on explicit depth limits and pre-defined capital allocation boundaries. If price movement exceeds the designed depth horizon without sufficient opportunity for partial relief or stabilization, the system transitions from absorption into saturation. Beyond this point, additional averaging no longer improves resilience but instead concentrates risk. This boundary is not probabilistic; it is structural.

Second, absorption deteriorates when capital pacing becomes misaligned with market tempo. The framework assumes that drawdowns unfold over time in a manner that allows incremental engagement, partial realization, and reconfiguration of exposure. In scenarios involving abrupt, near-vertical price collapses—where price traverses multiple depth levels faster than execution logic can respond—the system may be forced to deploy capital too rapidly, compressing the absorption window. In such cases, the failure mode is not theoretical ruin, but execution overload.

Third, absorption is compromised when liquidity conditions collapse. The absorption-behavior observed in this study presupposes the continued availability of spot liquidity sufficient to execute incremental trades without excessive slippage. In extreme market dislocations—such as exchange outages, severe order-book thinning, or sudden market-wide deleveraging—execution-level assumptions may fail. Under these conditions, the system's design objective shifts from absorption to preservation, often requiring manual intervention or suspension.

Finally, absorption ceases to function when structural discipline erodes. The framework is explicitly survival-first and depends on adherence to predefined execution rules, including pause conditions and maximum exposure thresholds.

Deviations motivated by discretion, impatience, or overconfidence effectively remove the structural safeguards that enable absorption in the first place. In this sense, failure is as often behavioral as it is mechanical.

These breakdown conditions underscore a central point: Martingale-derived absorption is not a free lunch. It operates within a bounded domain defined by depth, time, liquidity, and discipline. Outside this domain, the mechanism does not gradually degrade—it changes regime. Recognizing where absorption ends is therefore as critical as understanding how it functions.

By explicitly identifying these boundaries, the present analysis does not weaken the survivability argument. On the contrary, it clarifies that resilience in execution-level systems arises not from eliminating failure, but from knowing precisely where failure begins.

6. **Why Traditional Metrics Miss This Behavior**

Standard performance metrics commonly used in the evaluation of trading strategies—such as win rate, Sharpe ratio, or cumulative return curves—are poorly suited to capturing the behavior documented in this study. These metrics are designed to assess predictive efficiency, risk-adjusted returns, or distributional properties of returns, implicitly assuming that strategy performance can be summarized through stationary or expectation-based statistics.

The behavior observed in the depth-controlled Martingale system during prolonged drawdown episodes does not conform to these assumptions. The primary objective of the system is not to maximize return or optimize volatility-adjusted performance, but to maintain operational continuity under sustained adverse price paths. As a result, the most informative indicators of system behavior are not predictive statistics, but execution-level and structural metrics.

In this context, relevant evaluation dimensions include the maximum realized execution depth reached during stress, capital utilization as a function of drawdown progression, and the system's ability to remain active without structural reset. Equally important are the absence of liquidation or forced deleveraging events and the manner in which the system recovers following extreme adverse price paths. These indicators capture whether the system remains functionally intact, rather than whether it performs optimally in a statistical sense.

Such measures are more appropriately understood as engineering metrics rather than financial performance statistics. They describe how a system responds to stress, where its structural limits lie, and whether it transitions smoothly back to stable operation once pressure subsides. Traditional performance metrics may therefore misclassify structurally resilient systems as inefficient or suboptimal, precisely because they are blind to survivability, path dependence, and execution continuity.

From this perspective, evaluating Martingale-derived systems using conventional return-based metrics risks overlooking their most critical behavior: not how profits are generated, but how solvency is preserved when adverse price movements persist beyond the regimes for which predictive models remain valid.

7.Discussion

7.1 Role of Martingale Within a Diversified Strategy Set

It is important to clarify that the Martingale-based execution framework analyzed in this study does not represent the entirety of the author's investment strategy, nor is it intended to serve as a dominant source of portfolio returns. In practice, Martingale-based exposure accounts for less than 10% of total deployed capital, with the majority allocated to non-Martingale strategies exhibiting different risk profiles and return drivers.

Accordingly, the empirical observations reported here should not be interpreted as an endorsement of Martingale strategies as standalone or universally applicable solutions. Rather than illustrating survivorship bias or selective success, the present analysis reflects how a structurally constrained Martingale component behaves when embedded within a diversified strategy ensemble. Under such conditions, localized drawdowns or even complete failure of the Martingale module would not threaten overall portfolio solvency.

From a portfolio construction perspective, the findings suggest that Martingale-based mechanisms—when tightly bounded in depth, scale, and capital allocation—may function as peripheral risk absorption tools rather than primary performance engines. This framing aligns with broader principles of risk diversification, where individual components are deliberately designed to be non-critical, allowing investors to observe and exploit specific execution-level phenomena without exposing the entire portfolio to catastrophic risk.

7.2 Non-Overfitted Parameter Formation

It is important to emphasize that the parameter configuration of the analyzed Martingale system was not obtained through conventional backtesting optimization or retrospective curve fitting. Instead, the core structural parameters were initially formulated during the 2021–2022 cryptocurrency market crash through collective practitioner discussions within trading communities, reflecting real-time responses to extreme downside conditions rather than ex-post statistical tuning.

Following this initial formation, the parameter set underwent approximately one year of forward testing under live market conditions. During this period, adjustments were applied at the level of position sizing and depth allocation based on observed operational stress, without any retrospective recalibration or data mining across historical price series. Once finalized, the configuration remained fixed throughout the drawdown episode analyzed in this study.

Consequently, the results presented here do not represent a best-fit outcome derived from historical optimization, but rather the realized behavior of a structurally constrained system operating under unknown future market conditions. This forward-only evaluation framework directly addresses concerns raised in the literature regarding backtest overfitting and the limited external validity of performance-driven strategy studies.

7.3 On Strategy Evolution and Drawdown Absorption

This study does not seek to dispute the well-established concerns regarding overfitting and model fragility in quantitative trading research. Rather, it emphasizes that the primary object of analysis is not an optimized strategy, but the *evolutionary behavior of a trading system under prolonged adverse market conditions*.

The analyzed Martingale-based framework was developed with a singular objective: survivability during sustained drawdowns. Its structural features emerged through iterative adaptation to real market stress, particularly during prior market crashes, and were subsequently deployed without retrospective optimization. In this context, the observed outcome—namely, the system's ability to absorb a cumulative drawdown exceeding 40% while remaining operational and contributing positive cash flow—should not be interpreted as a claim of optimality or predictive power.

Instead, the result highlights an underexplored dimension of Martingale-derived systems: their potential role as *drawdown absorption mechanisms* within a diversified strategy set. From this perspective, the significance of the 40% drawdown episode lies not in outperforming the market, but in demonstrating how exposure can be progressively transformed and contained along realized adverse price paths, enabling system continuity rather than collapse.

Accordingly, the contribution of this work is not to rehabilitate Martingale strategies as universally effective tools, but to document how, under strict structural constraints and limited portfolio allocation, such strategies may participate in risk absorption rather than risk amplification during extreme downside regimes.

7.4 A Structural Challenge to Conventional Financial Economics

This paper does not claim that Martingale-based strategies are universally robust, nor does it attempt to rehabilitate Martingale systems as "safe" in any absolute sense. Such claims would be both misleading and intellectually dishonest. What this study demonstrates instead is more limited, yet more disruptive: Martingale-derived systems can be structurally re-engineered to prioritize survival over prediction, and when evaluated at the execution level rather than the price-process level, their behavior contradicts several core assumptions of conventional financial economics.

The ETH spot-market drawdown of approximately 42% serves as a concrete operational case. Under conditions widely considered hostile—deep, rapid, and prolonged downside movement — the strategy remained functional, capital-bounded, and non-failing. This outcome does not validate Martingale logic as a universal principle. It exposes a structural paradox: a strategy deemed theoretically fragile survives the realized market path, while many expectation-optimal strategies fail precisely because they cannot survive the path they were designed to optimize.

Crucially, this survivability is not accidental. The tracking Martingale mechanism employed in this study differs fundamentally from classical static Martingale schemes. Rather than mechanically doubling exposure, the tracking structure conditionally deploys capital in response to market displacement and partial recovery, allowing the system to repeatedly engage at relatively lower local price levels. By doing so, it mitigates one of the core weaknesses of classical Martingale strategies—poor average entry quality during extended trends—while preserving their ability to absorb downside movement.

During the ETH drawdown episode, this tracking behavior allowed the system to progressively improve cost basis through multiple localized engagements, effectively dampening the net impact of continued price declines. In operational terms, repeated partial absorptions of downside pressure reduced effective drawdown severity, transforming a large directional move into a sequence of manageable execution events. The drawdown was not avoided; it was *absorbed*.

This paradox does not arise from statistical error, curve fitting, or mismeasurement. It arises from a mismatch between what financial economics models and what markets actually demand.

Modern financial economics optimizes expectation under assumed probability distributions. Markets, by contrast, enforce survival constraints. Particularly in cryptocurrency markets—characterized by non-stationarity, fat tails, regime shifts, and frequent structural breaks—average outcomes are largely irrelevant. What matters is whether a system remains operational when the distributional assumptions fail. As Taleb (2007) emphasizes, exposure to tail risk is not about the size of losses in expectation, but about whether the system remains alive once the tail arrives.

The depth-controlled tracking Martingale framework examined here does not attempt to eliminate drawdowns or disguise risk through smoothing. Instead, it treats drawdowns as structural inputs that activate bounded, pre-engineered execution responses. From the perspective of probabilistic rationality, such behavior appears irrational. From the perspective of survivability, it is internally coherent. The contradiction lies not in the strategy, but in the evaluative framework applied to it.

Probability-based risk metrics—variance, Value-at-Risk, Sharpe ratios—implicitly assume ergodicity, stationarity, and temporal independence. Cryptocurrency markets violate all three. As Peters (2019) argues, in non-ergodic systems, ensemble averages are not meaningful indicators of individual system outcomes. What matters is time-average growth and the avoidance of absorbing barriers such as liquidation or capital exhaustion. In this context, the relevant question is not whether expected return is positive, but whether the strategy remains alive after the worst plausible path. Classical finance offers no adequate language for this question.

Martingale strategies are routinely dismissed in literature as a single pathological class associated with reckless doubling and inevitable ruin. This treatment collapses a critical distinction between unbounded Martingale escalation and bounded, depth-limited tracking Martingale decomposition. The former is indeed pathological. The latter, as demonstrated in this study, functions as a nonlinear cost-basis adjustment mechanism operating under explicit survival constraints. By failing to distinguish between these forms, mainstream economics discards an entire class of execution-level designs without empirical discrimination.

A further source of misunderstanding lies in the level of abstraction. Financial economics largely operates at the price-process level, where execution dynamics are treated as secondary frictions or ignored altogether. Traders, however, operate at the execution level, where survival is governed by margin mechanics, liquidity availability, capital pacing, and exhaustion. The strategy examined here does not predict prices; it engineers responses to price movement. This places it outside the conceptual vocabulary of traditional asset pricing theory, but squarely within the operational reality of real markets.

The resistance of conventional peer review to this framework is therefore unsurprising. The challenge posed by this work is not to a specific model, but to the dominant evaluative paradigm itself. Cryptocurrency markets may require a reframing of what constitutes a valid trading system—one in which robustness, boundedness, and survivability precede equilibrium, efficiency, or expected alpha. This does not invalidate financial economics; it delineates its domain limits.

The ETH drawdown episode does not demonstrate that Martingale strategies are safe. It demonstrates something more unsettling: certain strategies survive precisely because they violate the assumptions used to condemn them. This is not a contradiction to be resolved through improved statistics or more refined risk metrics. It is a signal that the model and the market are solving fundamentally different problems.

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8. Concluding Remark

The 42% ETH drawdown described here is not a backtest artifact, nor a simulated stress scenario. It is a real market event encountered with real capital. The strategy did not outperform the market by avoiding the drawdown; it endured it. In markets where extreme events are the norm rather than the exception, this distinction may matter more than any single performance metric.

By documenting this episode at the execution level, this study does not argue for the universal adoption of Martingale-based strategies. Instead, it highlights the importance of evaluating trading systems through survivability, structural boundaries, and response to realized stress paths rather than outcome-based optimality alone.

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Declaration of Interest

The author declares that there are no competing interests associated with this manuscript.

Data Availability Statement

The data used in this study consist of proprietary trading records and are not publicly available due to privacy and operational constraints. Aggregated statistics and reconstructed path-level summaries supporting the findings of this study are provided within the manuscript and the appendix.

Use of Generative AI

Generative AI tools were used in the preparation of this manuscript to assist with English language refinement and clarity of expression. Specifically, ChatGPT (OpenAI, GPT-4 series) was used to improve grammar, sentence flow, and academic tone.

The AI tools were not used to generate research ideas, data, analyses, or conclusions. All substantive content, interpretations, and conclusions are the sole responsibility of the author.

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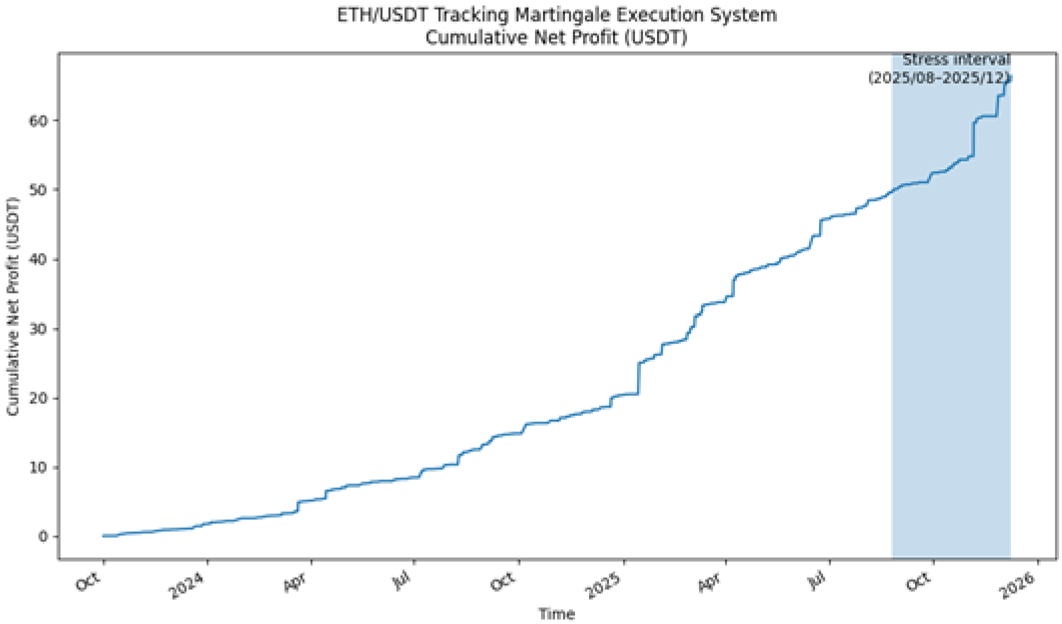

Figure 1. Cumulative net profit (USDT) of the ETH/USDT spot tracking-Martingale execution system over live operation.

The curve is constructed by cumulatively summing the realized profit entries from the exchange-generated execution log. The shaded region highlights the prolonged downside stress interval from **August 25, 2025 to December 18, 2025**, corresponding to the drawdown event analyzed in this study.

Figure 2. ETH spot price trajectory during the prolonged drawdown shock (August 25, 2025–December 18, 2025).

The figure illustrates the ETH/USDT spot price path from August 2025 to December 2025. The upper horizontal reference line (A) marks the local price peak near 4,950 USDT, while the lower reference line (B) indicates the subsequent trough around 2,830 USDT reached during the market-wide sentiment collapse. The peak-to-trough decline from A to B corresponds to a cumulative drawdown of approximately 42.8%.

The shaded region highlights the prolonged downside stress interval analyzed in this study, characterized by sustained selling pressure, intermittent relief rallies, and repeated breakdowns of intermediate support levels. This interval constitutes the real-market stress environment within which the execution-level behavior of the ETH spot tracking-Martingale system is evaluated. The same time window is highlighted as the shaded region in Figure 1, enabling a direct, time-aligned comparison between exogenous market drawdown dynamics and endogenous execution-level system behavior.